

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
4th Semester of B. Pharm. Examination
University Theory Examination October/November 2017
PH 212 Physical Pharmaceutics

Date: 05/10/2017, Thursday Time: 01:30 p.m. to 04:30 p.m. Maximum Marks: 80

Instructions:

- 1. Figures to right indicate marks.**
- 2. Section wise separate answer book to be used.**
- 3. Draw neat sketches wherever necessary.**

SECTION- I

Q 1 Attempt any TWO of the following

Give the various methods for determination of particle size. Briefly explain

A principle, working, advantages and disadvantages of optical microscopy method. 10

Give the difference between Newtonian & non Newtonian flow. Describe

B principle, working, application, advantages and disadvantages of cone & plate viscometer. 10

Give and define derived properties of Powders. Briefly explain the importance of determination of porosity of powders in pharmacy.

C 10

Q 2 Attempt any TWO of the following

Define sublimation, eutectic mixtures, liquid complexes, amorphous and polymorphism. Give the application of X Ray Diffraction (XRD) and Fourier Transform Infra-Red Spectroscopy (FTIR) for characterization of polymorphs.

A 10

Give the various methods for measurement of surface and interfacial tensions.

B Briefly explain principle, working, advantages and disadvantages any one suitable method. . 10

Describe types of non-Newtonian flow. How the thixotropy can be measured? Explain.

C 10

SECTION- II

Q 3 Attempt any TWO of the following

- A Give definition, types, properties and application of colloids in Pharmacy. 10
Briefly describe protective colloids.
Describe the parameters for physical instability of an Emulsion. How does
- B electrokinetic properties and globule size may affect the physical stability of 10
Emulsion?
- C Give definition, types, properties and application of suspension in Pharmacy. 10
Differentiate flocculated suspension with deflocculated suspension.

Q 4 Attempt any TWO of the following

- A Give types, theory and application of Emulsion in Pharmacy. 10
- B Give classification, methods of preparation and applications of complexes. 10
- C Define Buffer solution and Isotonic solution and Give their pharmaceutical 10
applications. Describe the methods for adjusting tonicity.
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